HIGHER DIRECT IMAGES OF A TORSION SHEAF, AND OF A COMPLEX WITH TORSION COHOMOLOGY, ALONG A QUASI-COMPACT QUASI-SEPARATED MORPHISM ARE TORSION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full for both parts: the presheaf description of the higher direct images, the reduction to affine V where the fibre product is quasi-compact and quasi-separated, and the appeal to the torsion statement.

Editorial note. The label lemma-torsion-direct-image names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Description of the higher direct images by the cohomology of the base change. Cohomology on Sites, the lemmas identifying R^n f_* of a sheaf and of a complex with the sheaf associated to V |-> H^n(X x_Y V, -). Supplies the presheaf description used to reduce both assertions to a computation on X x_Y V.
- Torsion cohomology of a quasi-compact quasi-separated scheme. Same chapter, the lemma stating that the etale cohomology of a torsion sheaf on a quasi-compact quasi-separated scheme is torsion. Supplies the torsion property of the cohomology groups appearing in the presheaf description.

Source reference. Original reference 0DDC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 13899–13899. Prerequisite pointer invoked at proof line(s) 13943; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 072W: sources/stacks-chow/tags.txt, source lines 9040–9040. Prerequisite pointer invoked at proof line(s) 13940; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D6I: sources/stacks-chow/tags.txt, source lines 16523–16523. Prerequisite pointer invoked at proof line(s) 13950; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of schemes.*

(1) *If $\mathcal{F}$ is a torsion abelian sheaf on $X_{étale}$, then $R^n f_* \mathcal{F}$ is a torsion abelian sheaf on $Y_{étale}$ for all n.*
(2) *If K in $D^+(X_{étale})$ has torsion cohomology sheaves, then Rf_*K is an object of $D^+(Y_{étale})$ whose cohomology sheaves are torsion abelian sheaves.*

Proof. Proof of (1). Recall that $R^n f_* \mathcal{F}$ is the sheaf associated to the presheaf $V \mapsto H^n_{étale}(X \times_Y V, \mathcal{F})$ on $Y_{étale}$. See Cohomology on Sites, Lemma 072W. If we choose V affine, then $X \times_Y V$ is quasi-compact and quasi-separated because f is, hence we can apply Lemma 0DDC to see that $H^n_{étale}(X \times_Y V, \mathcal{F})$ is torsion.

Proof of (2). Recall that $R^n f_* K$ is the sheaf associated to the presheaf $V \mapsto H^n_{étale}(X \times_Y V, K)$ on $Y_{étale}$. See Cohomology on Sites, Lemma 0D6I. If we choose V affine, then $X \times_Y V$ is quasi-compact and quasi-separated because f is, hence we can apply Lemma 0DDC to see that $H^n_{étale}(X \times_Y V, K)$ is torsion. □